



THE IMPACT OF HOUSEHOLD INCOME ON COVID-19 DEATHS

Tzu-Han Lin, Ching-yu Huang

School of Computer Science and technology, Kean University, Union, NJ, USA

{lintzuh, chuang}@kean.edu

Abstract

Covid-19 threats global since 2020, there are still many hidden information about how demographic environment interacts with the Covid-19 disease. By joining the 2020 US Census and Covid-19 datasets in the database, we were able to analyze many attributes and found that the Covid-19 deaths is significantly related to the household income.

Datasets and Methods

1. Two different datasets: 2020 U.S. Census and Covid-19 stored in 2 different MySQL tables.
2. 2935 US counties and Covid-19 data from 1/21/2020 to 12/31/2021 were used for the analysis.
3. Use google charts to visualize the datasets and results.
4. Use SQL to join these two tables and set the condition to eliminate the non-updating data and small population area.

Platforms

- Linux system, MySQL Database
- Code: HTML, JavaScript, PHP, Python
- API: Google Charts

References

- 2020 US Census dataset: <https://simplemaps.com/>
- Covid-19 dataset: <https://covid.ourworldindata.org/>

state	county	deaths 2021_12	cases 2021_12	income household median	death rate
Texas	Sabine	77	986	34992	7.8093
Georgia	Hancock	83	1210	31860	6.8595
Texas	Knox	22	390	48798	5.641
South Dakota	Jerauld	19	337	49632	5.638
Georgia	Candler	90	1619	33736	5.559
Texas	Newton	54	987	40101	5.4711
Texas	Haskell	37	686	40313	5.3936
Texas	Dickens	15	280	42540	5.3571
Georgia	Twiggs	62	1168	34879	5.3082
Georgia	Jenkins	63	1210	27375	5.2066

Join two table using MySQL workbench.

Column description:

cases2021_12: accumulation the total cases of Covid-19 from 01/21/2020 to 12/31/2021

population: population in the county

deaths2021_12: accumulation the total death cases of Covid-19 from 01/21/2020 to 12/31/2021

income_household_median: Median household income in USD.

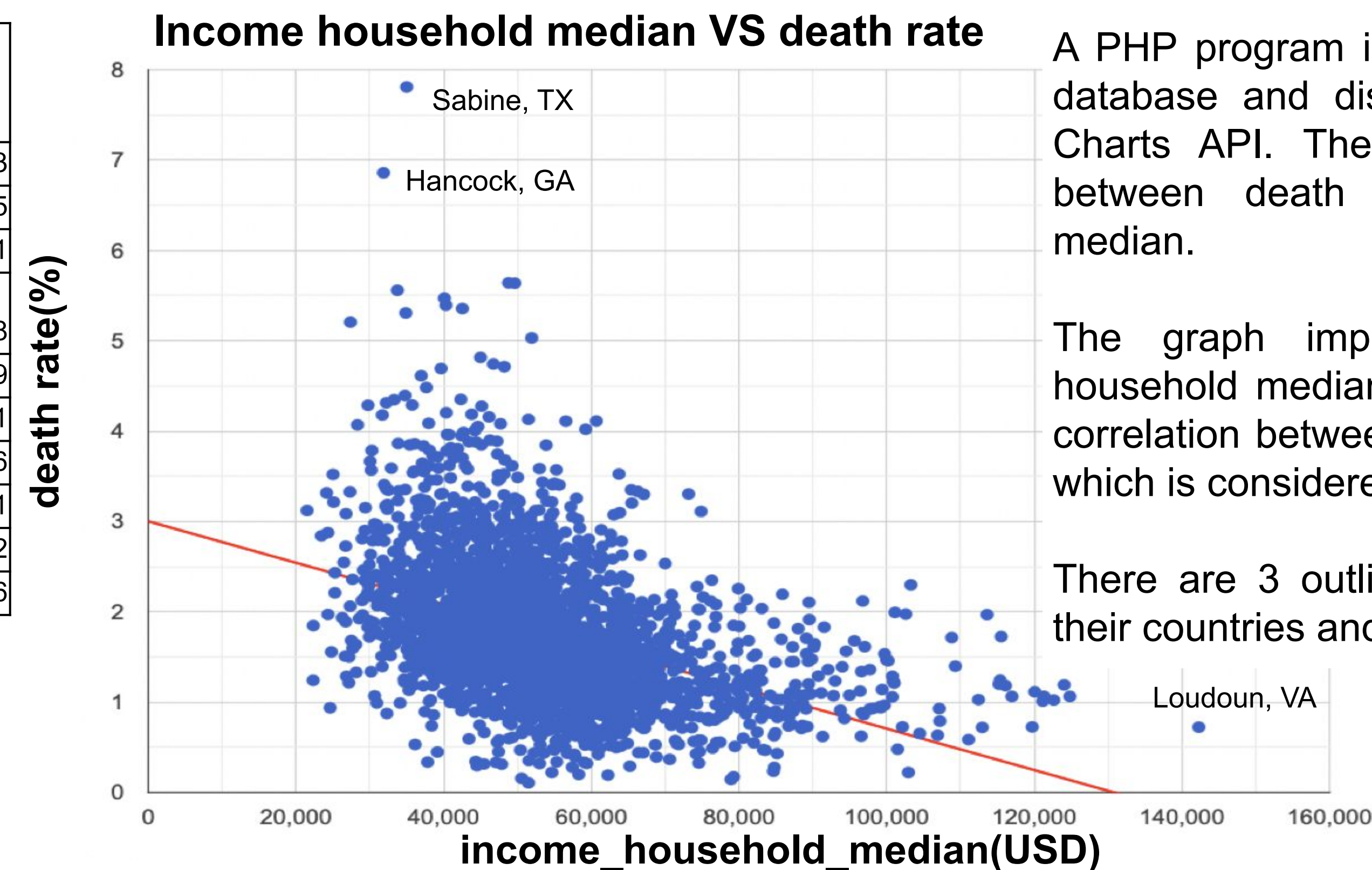
death_rate=death numbers/ case numbers

The top 10 high death rate counties are mostly at Texas and Georgia.

	Q1	Q2	Q3	max
income median	44220	51745	59710	142299
deaths Total	78219	152018	174640	371909
cases Total	3735026	8645996	11614344	27617709
expected Deaths	56212.80	130123.78	174797.95	415651.45

Divide 2935 counties into 4 groups base on their income median quartile and operate the chi-square test. Null hypothesis: deaths number is not dependent with income household median degree of freedom:3, chi-square = 16905.79 p-value<<0.00001

Experiments and Results



A PHP program is access the data from MySQL database and display the results using Google Charts API. The chart shows the relationship between death rate and income household median.

The graph implies that the higher income household median has the lower death rate. The correlation between those two variables is -0.425, which is considered a weak relation.

There are 3 outliers indicated on the chart with their countries and states.

county	deaths 2021_12	cases 2021_12	death rate(%)	population	density	age median	income household median
Sabine	77	986	7.8093	10471	8.22	53	34992
Jasper	165	4252	3.8805	35506	14.6	40.5	44370
Shelby	114	3077	3.7049	25349	12.3	38	42522
Newton	54	987	5.4711	13914	5.75	44.3	40101
San Augustine	38	904	4.2035	8286	6.03	49.3	40353

county	deaths 2021_12	cases 2021_12	death rate(%)	population	density	age median	income household median
Hancock	83	1210	6.8595	8515	6.98	45.6	31860
Baldwin	203	7049	2.8798	45111	67.33	35.5	43672
Putnam	89	3497	2.545	21691	24.3	46.2	51606
Washington	126	3326	3.7883	20436	11.63	41.5	38068
Greene	74	3008	2.4601	17349	17.28	50.2	52129
Warren	26	809	3.2138	5297	7.19	45.1	37203
Glascock	21	436	4.8165	2996	8.05	42.7	45000

In both the counties nearby Sabine, TX and Hancock GA, Sabine county and Hancock county both have the lowest income median which agree with our result. Furthermore, Greene county has the highest income median which indeed has the lowest death rate

Summary

By scatter plot, we found that there is a negative correlation between income household median and the Covid-19 deaths. Also, the chi square test result reject the hypothesis that the deaths number is not dependent with the income median.

In addition, the result of analyzing the adjacent counties of the outliers at the scatter plot agree with our previous observation.

We infer people in the county which has higher income household median maybe have more ability to afford medical treatment for Covid-19.

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